

LARRY HUI

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EDUCATION

M.S. in Robotics, University of Pennsylvania (GRASP Lab)

Expected May 2028

Coursework: World Models, Control and Optimization for Robotics, Machine Perception, Distributed Robotics

B.S in Mechanical Engineering (EECS focus), University of California, Berkeley

May 2026

GPA: 3.59/4.0 (last two years: 3.91/4.0) | 2× Dean's Honors List (Top 10%)

Coursework: Robotics, Bipedal Robot Control, Convex Optimization, Machine Learning, Feedback Controls, Mechatronics, Deep RL for Controls, Embedded Systems, Continuum Mechanics, FEA

EXPERIENCE

Mechanical Engineering Intern, Outer Rim Exploration Inc., Tokyo, Japan

Jul 2024 – Dec 2024

- Designed the CAD assembly for a drone-mounted muon hodoscope, integrating PCB layouts and validation hardware.
- Designed a geared, solenoid-locked rotation stage that sets the detector's zenith angle to degree-level accuracy.
- Reduced mass by 10% and increased stiffness via geometric optimization, verified with custom MATLAB FEA solver.
- Cut prototype cost by 40% with COTS parts, helping secure \$200K in seed funding and a first pilot contract.

Undergraduate Researcher, Gu Research Group, UC Berkeley

Dec. 2024 – May 2026

- Built PINN and physics-informed MeshGraphNet solvers for linear-elasticity PDEs in MATLAB and PyTorch for a MathWorks-sponsored project, using ensembles, a **192-mesh** training pool with metadata, and cosine annealing.
- Benchmarked PINN ensembles against FEA across 3 geometries (Timoshenko beam, plate with central hole, clamped plate), 3 BC types, and 4 loadings, achieving **<1.3%** displacement error over **15k+ collocation points**.
- Built a differentiable GNN solver for fluid–solid interaction with normalized input/output features and LayerNorm; a fluid-only warm start cut flap-displacement error from **7–32%** to **<5.2%** on unseen geometries.

Lead Teaching Assistant, Mechanical Engineering, UC Berkeley

May 2025 – May 2026

- Led ME 103 Measurements (180+ students) and ME 132 Feedback Control (80+), coordinating 8 TAs and readers.
- Created labs & exams on SP, controller design with Root Locus, FR, and state-space in LabVIEW, MATLAB, Simulink.

PROJECTS

Cable-Actuated Air Hockey Robot

Jan. 2026 – May 2026

- Co-designed a 4-BLDC-motor cable-driven robot on an 80/20 frame, minimizing moving inertia for fast mallet motion.
- Developed a RT-MPC planner with quintic-spline trajectories & PPO policy scoring 83% of shots vs. 29% for Q-learning.
- Tracked the puck to **<0.1 mm** by fusing homography- and parallax-corrected OpenCV detections in an EKF.
- Derived closed-form IK and least-squares FK, streaming closed-loop commands to moteus motor controllers over CAN.

Autonomous Cable Insertion with UR7e Arm

Aug. 2025 – Dec. 2025

- Built a ROS2 IK/CV-control stack that localizes cables and ports with RealSense depth and detects USB-C, achieving mm-level insertion accuracy across randomized trials.
- Designed a custom end-effector and force limited-compliant insertion with retry/re-grasp recovery from misalignment.

Pixel Mirror: RT LED Display with Virtual Gloving

Jan. 2026 – May 2026

- Achieved <70 ms end-to-end latency at 30 fps by porting a YOLOv8-seg silhouette pipeline to native C++; streamed frames over a 1 Mbaud serial link to ESP32 firmware that drives LED board via I2S DMA with minimal CPU overhead.

RL for Humanoid Locomotion on Rough & Transient Terrain

Aug. 2025 – Dec. 2025

- Trained a Unitree G1 walking policy in IsaacLab via PPO (2,048 parallel envs) on a rough-terrain curriculum.
- Implemented BeamDojo's double-critic PPO from scratch; built moving stepping-stone terrain in MuJoCo MJX.

PUBLICATIONS

1. **L. Hui**, J. Kam, W. Su, and J. Zhou, "Hamilton–Jacobi Reachability for Spacecraft Collision Avoidance," *2026 IEEE International Conference on Control & Automation (ICCA)*, 2026.

SKILLS AND INTERESTS

CAD & Simulation

Languages/Frameworks

Libraries/Tools/OS

Hardware/Embedded:

Languages/Interests

SolidWorks, Fusion360, OnShape, ANSYS, ABAQUS, Simulink, Gmsh, GD&T, DFM

Python, ROS2, PyTorch, NVIDIA Isaac, C++, MATLAB, Java, Git, LabVIEW

NumPy, SciPy, Pandas, MuJoCo, CVX, Linux (Ubuntu), macOS, FreeRTOS

ESP32, Jetson Nano, CAN, UART, I2S/DMA, Machining, 3D Printing, Instron

English, Mandarin, French (B2); Private Pilot (FAA); ARCT piano diploma